

Research themes

Intelligence, Computing & Communication Lab

THEME 01

Wireless communication

- 5G/6G and radio access networks
- MIMO and antenna arrays
- Cognitive radio and spectrum sharing
- Uplink, downlink and relaying
- Aerial and backscatter communication

THEME 02

Edge computing and IoT

- Internet of Things (IoT)
- Mobile edge computing
- Sensor networks and sensing
- Network slicing and virtualization
- Real-time processing and services

THEME 03

Artificial intelligence

- Machine and deep learning
- Reinforcement learning
- Neural networks and transformers
- Feature extraction and training
- Data-driven computational modelling

THEME 04

Optimization and resource allocation

- Power and bandwidth allocation
- Scheduling and admission control
- Energy efficiency and harvesting
- Throughput–latency trade-offs
- Convex approximation and heuristics

THEME 05

Signal processing and inference

- Array signal processing and precoding
- Channel estimation and decoding
- Interference and noise mitigation
- Radar and integrated sensing
- Statistical modelling and inference

THEME 06

Security and reliability

- Communication-system security
- Jamming and eavesdropping
- Privacy and encryption
- Reliable low-latency communication
- Robustness under uncertainty